



INTERNSHIP REPORT PRESENTATION

Report Of Evaluation of AI Software Developer Intern at Trovex.ai

Name: Yash Mahindroo

Roll No.: CS-2341159

Section: 4CSE-1

Batch: 2023 - 2027

Program: B.Tech CSE (AI/ML)

University: IILM University, Greater Noida

Organization: Trovex.ai

Role: AI Software Developer Intern

About the Company

Trovex.ai is an AI-powered sales enablement and training company, founded in 2023 and headquartered in Bengaluru, India.

- Uses AI-driven roleplay simulations to help sales and customer support teams practice real-world conversations.
- Provides personalized AI feedback on pitching, objection-handling, and negotiation skills.
- Engineering workflow relies heavily on large language models — for both product features and internal development processes such as automated test-case generation.
- Actively investing in AI infrastructure — knowledge graphs, retrieval pipelines, and evaluation tooling — to support and scale its product.

Internship Overview

As an AI Software Developer Intern, I worked across four connected workstreams supporting Trovex.ai's AI infrastructure and product quality:

- RAG Retrieval Evaluation — Benchmarked reranking models to improve the accuracy of the knowledge-graph-based retrieval pipeline.
- QA Automation Research — Evaluated an AI-driven, no-code testing tool as a potential addition to the team's QA workflow.
- Prompt Governance — Took ownership of naming conventions and folder structure for prompt assets in the observability platform.
- Manual QA Support — Performed manual functional testing of new features before promotion to production.

Problem Statement

The team needed a data-driven way to decide how to improve retrieval quality in its RAG pipeline, and whether existing manual QA processes could be safely and effectively automated.

- No prior comparison existed between a managed cloud reranking service and open-source alternatives for the knowledge graph.
- QA relied entirely on manual test execution — no evaluation of automation tools had been done.
- Prompt assets in the observability platform lacked a consistent naming and folder convention, causing maintainability issues as usage grew.

Project Objectives

- Evaluate and benchmark a managed cloud reranking service against open-source reranking models for retrieval accuracy.
- Diagnose and document data quality issues in the production knowledge graph affecting evaluation validity.
- Assess the feasibility of an AI-driven, no-code QA automation tool for the team's testing workflow.
- Establish a naming convention and folder structure for prompt assets in the observability platform.
- Support the team's QA process through manual testing of new features prior to release.

Tools & Technologies

Neo4j: Graph database storing the knowledge graph; used for schema inspection and vector similarity search.

Google Cloud Vertex AI: Text embeddings (Gemini) and the managed Ranking API used for reranking evaluation.

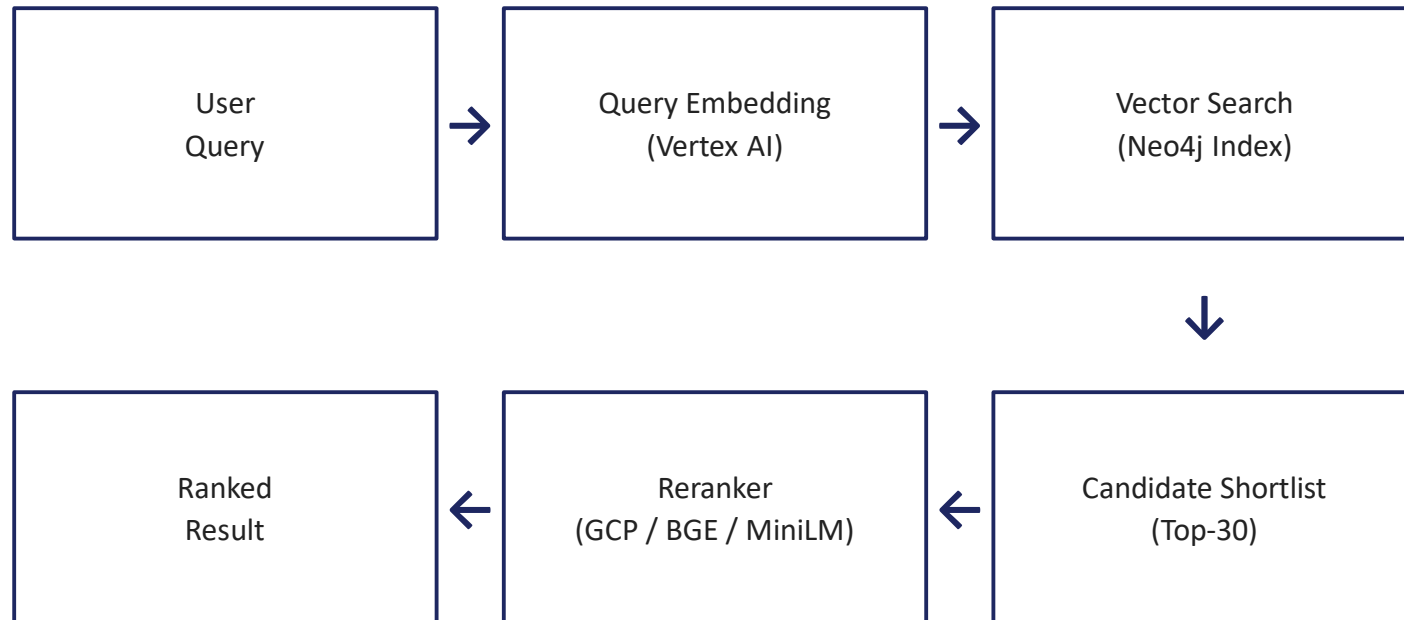
Python: Built the evaluation pipelines using Neo4j and Google Cloud client libraries.

BGE / MiniLM: Open-source cross-encoder rerankers run locally as a baseline comparison.

Momentic.ai: No-code AI QA automation tool evaluated for the team's testing workflow.

Langfuse: LLM observability platform; owned prompt naming and folder conventions here.

System Architecture — Retrieval Pipeline

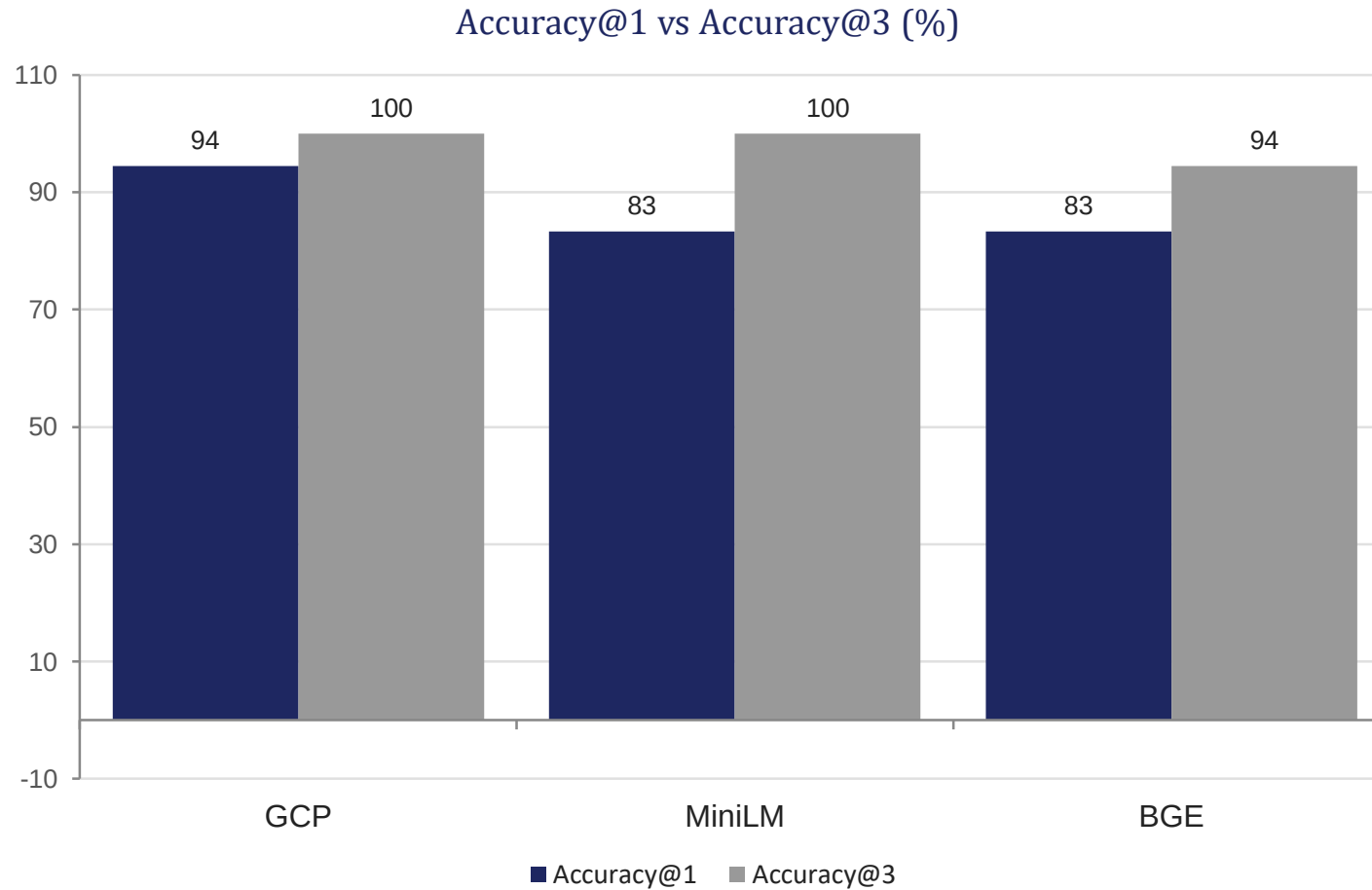


Flow: Query is embedded, matched against the knowledge graph via vector search, narrowed to a top-30 shortlist, then reranked for final relevance ordering.

Methodology

- Controlled Comparison — Same 18 evaluation questions and identical top-30 candidate pool used across all three rerankers, so reranker was the only variable.
- Ground Truth Mapping — Each question manually mapped to its correct source document to enable objective scoring.
- Standard IR Metrics — Scored using Accuracy@1, Accuracy@3, and average response time per query.
- Data Quality Audit — Knowledge graph audited for duplicate documents and folder inconsistencies before evaluation.

Results — Reranker Comparison



94.44% — GCP Reranker

Accuracy@1, highest of the three

1.87s — GCP average response time
per query (API)

2 / 18 — Questions where GCP
outperformed both local models

Other Contributions

- QA Automation Evaluation — Set up and trialed Momentic.ai against a staging environment; adoption paused pending management approval and a credential-security review.
- Prompt Governance (Langfuse) — Took ownership of the prompt folder structure, renamed existing prompts, and defined a naming convention for future assets.
- Manual QA Testing — Performed manual functional testing of new features on the staging environment prior to production promotion.

Key Learnings

- Data quality precedes evaluation — Duplicate documents and inconsistent folder structures had to be resolved before any reranker comparison could be considered valid.
- Isolating failure stages matters — Distinguishing retrieval-stage failures from reranking-stage failures was essential to correctly interpret evaluation results.
- Cloud IAM is its own skill — Diagnosing permission and API configuration issues on Google Cloud required systematic elimination of possible causes.

Conclusion

This internship delivered a measurable, data-backed recommendation on reranking strategy, a clearer picture of the team's data quality practices, and tangible improvements to QA and prompt management processes at Trovex.ai.

Thank You

Yash Mahindroo | CS-2341159 | 4CSE-1 | Batch 2023-2027
IILM University | B.Tech CSE (AI/ML) | Trovex.ai